

Digital Image Processing Using MATLAB

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Road Map

- Array versus Matrix Operations
- Linear Versus Non-Linear Operations
- Arithmetic Operations
- Logical Operations
- Image Transformation

Array versus Matrix Operation

➤ Array Operations

Let f and g are two 2-Dimensional arrays. Four possible operations in are :

Addition, Subtraction, Multiplication and division

$$a = f(x, y) + g(x, y)$$

$$s = f(x, y) - g(x, y)$$

$$m = f(x, y) .* g(x, y)$$

$$d = f(x, y) ./ g(x, y)$$

➤ Matrix Operations are Multiplication and division

$$a = f(x, y) * g(x, y)$$

$$s = f(x, y) / g(x, y)$$

Linear Versus Non-Linear Operations

- Consider a general operator H , that produces an output image $g(x, y)$ from given input image $f(x, y)$:

$$H[f(x, y)] = g(x, y)$$

- Operator H is said to be a linear if

$$H[a * f_1(x, y) + b * f_2(x, y)] = a * H[f_1(x, y)] + b * H[f_2(x, y)]$$
$$a * g_1(x, y) + b * g_2(x, y)$$

Here, a and b are constants.

Linear Versus Non-Linear Operations

- Σ is a linear operator
- Min and Max are not linear operator

$$\max(x, x_0) = \begin{cases} x_0 & x < x_0 \\ x & x \geq x_0 \end{cases}$$

So you have a function that is constant for some value and linear for other. So is it possible that $\max(x, x_0) = x_0$ and $\max(y, x_0) = x_0$, but $\max(y + x, x_0) = x + y$ that is not equal to the $x_0 + x_0$. This contradicts linearity.

Arithmetic Operations

➤ Addition

For any two digital images, say f and g , the addition would yield the output image z with the arithmetic addition of the pixel values of f and g at the (x, y) coordinates respectively.

➤ Subtraction

For any two digital images, say f and g , the addition would yield the output image z with the arithmetic addition of the pixel values of f and g at the (x, y) coordinates respectively.

➤ Multiplication

For any two digital images, say f and g , the addition would yield the output image z with the arithmetic addition of the pixel values of f and g at the (x, y) coordinates respectively.

➤ Division

For any two digital images, say f and g , the addition would yield the output image z with the arithmetic addition of the pixel values of f and g at the (x, y) coordinates respectively.

Addition Operation

IMAGE 1

0	0	0	0
0	100	100	0
0	100	100	0
0	0	0	0

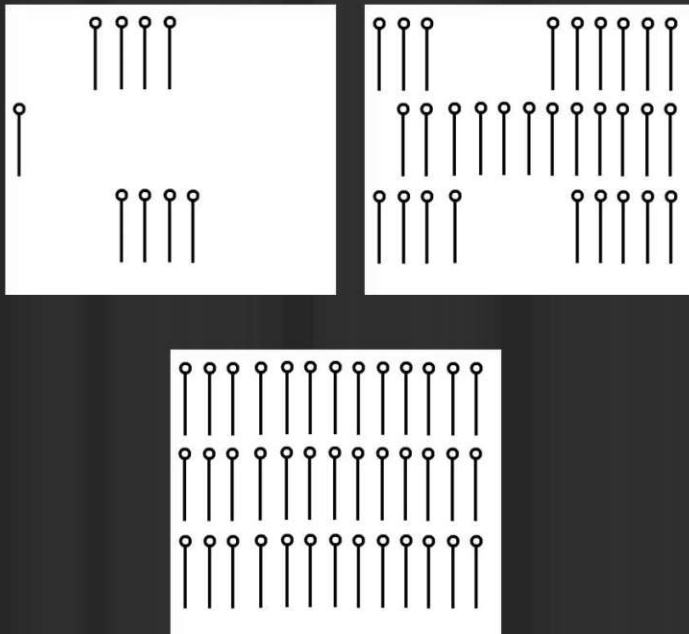
IMAGE 2

100	0	0	0
0	100	0	0
0	0	100	0
0	0	0	100

IMAGE 1 + IMAGE 2

100	0	0	0
0	200	100	0
0	100	200	0
0	0	0	100

Arithmetic Operation (Addition)



```
clear;  
clc;  
img1=imread('D:\Imageprocessing\IP_sttp\sticks3.bmp');  
img2=imread('D:\ImageProcessing\IP_sttp\sticks2.bmp');  
img1=double(img1);  
img2=double(img2);  
imshow(img1);  
figure,imshow(img2);  
final=img1 + img2;  
final=mat2gray(final);  
imwrite(final,'D:\Image Processing\IP_sttp\sticks.bmp');  
figure, imshow (final);
```

Subtraction Operation

IMAGE 1

0	0	0	0
0	100	100	0
0	100	100	0
0	0	0	0

IMAGE 2

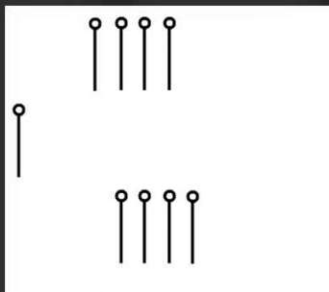
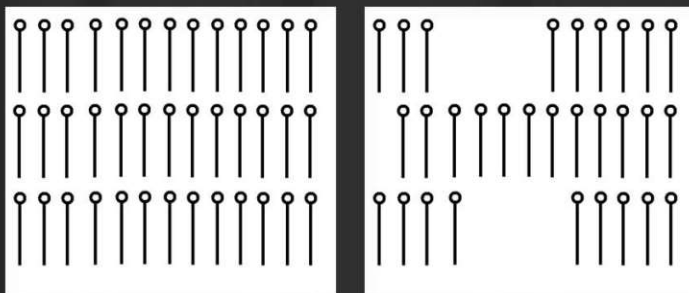
100	0	0	0
0	100	0	0
0	0	100	0
0	0	0	100

IMAGE 1 - IMAGE 2

0	0	0	0
0	0	100	0
0	100	0	0
0	0	0	0

Image subtraction is a basic tool in medical imaging, where it is used to remove static background information

Arithmetic Operation (Subtraction)



```
clear;  
clc;  
img1=imread('D:\Image processing\IP_sttp\sticks.bmp');  
img2=imread('D:\Image processing\IP_sttp\sticks2.bmp');  
img1=double(img1);  
img2=double(img2);  
imshow(img1);  
figure,imshow(img2);  
final=img1 -img2;  
final=mat2gray(final);  
imwrite(final,'D:\Image Processing\IP_sttp\sticks3.bmp');  
figure, imshow (final);
```

Multiplication Operation

IMAGE 1

0	0	0	0
0	100	100	0
0	100	100	0
0	0	0	0

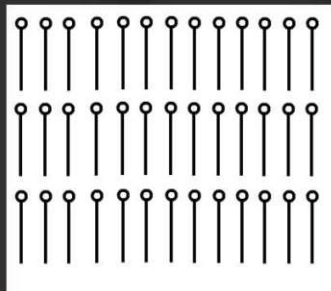
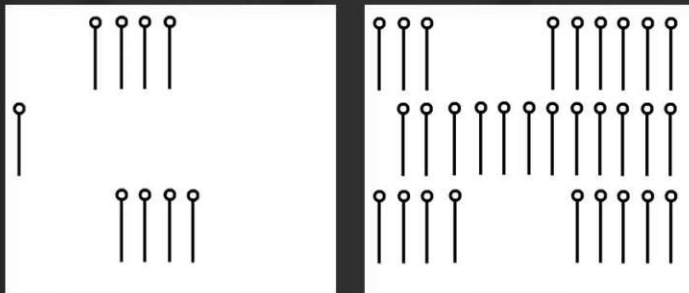
IMAGE 2

100	0	0	0
0	100	0	0
0	0	100	0
0	0	0	100

IMAGE 1 * IMAGE 2

0	0	0	0
0	255	0	0
0	0	255	0
0	0	0	0

Arithmetic Operation (Multiplication)



```
clear;  
clc;  
img1=imread('D:\Imageprocessing\IP_sttp\sticks3.bmp');  
img2=imread('D:\ImageProcessing\IP_sttp\sticks2.bmp');  
img1=double(img1);  
img2=double(img2);  
imshow(img1);  
figure,imshow(img2);  
final=img1 .* img2;  
final=mat2gray(final);  
imwrite(final,'D:\Image Processing\IP_sttp\sticks.bmp');  
figure, imshow (final);
```

Division Operation

IMAGE 1

0	0	0	0
0	100	100	0
0	100	100	0
0	0	0	0

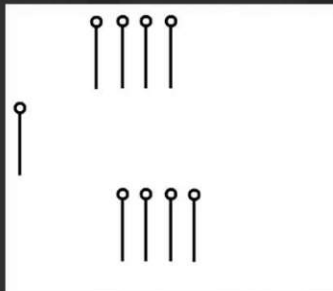
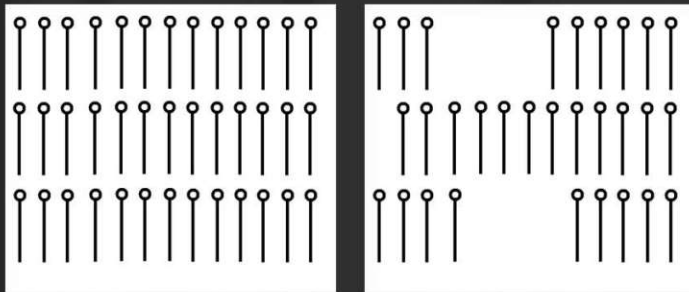
IMAGE 2

100	0	0	0
0	100	0	0
0	0	100	0
0	0	0	100

IMAGE 1 / IMAGE 2

0	0	0	0
0	1	0	0
0	0	1	0
0	0	0	0

Arithmetic Operation (Division)

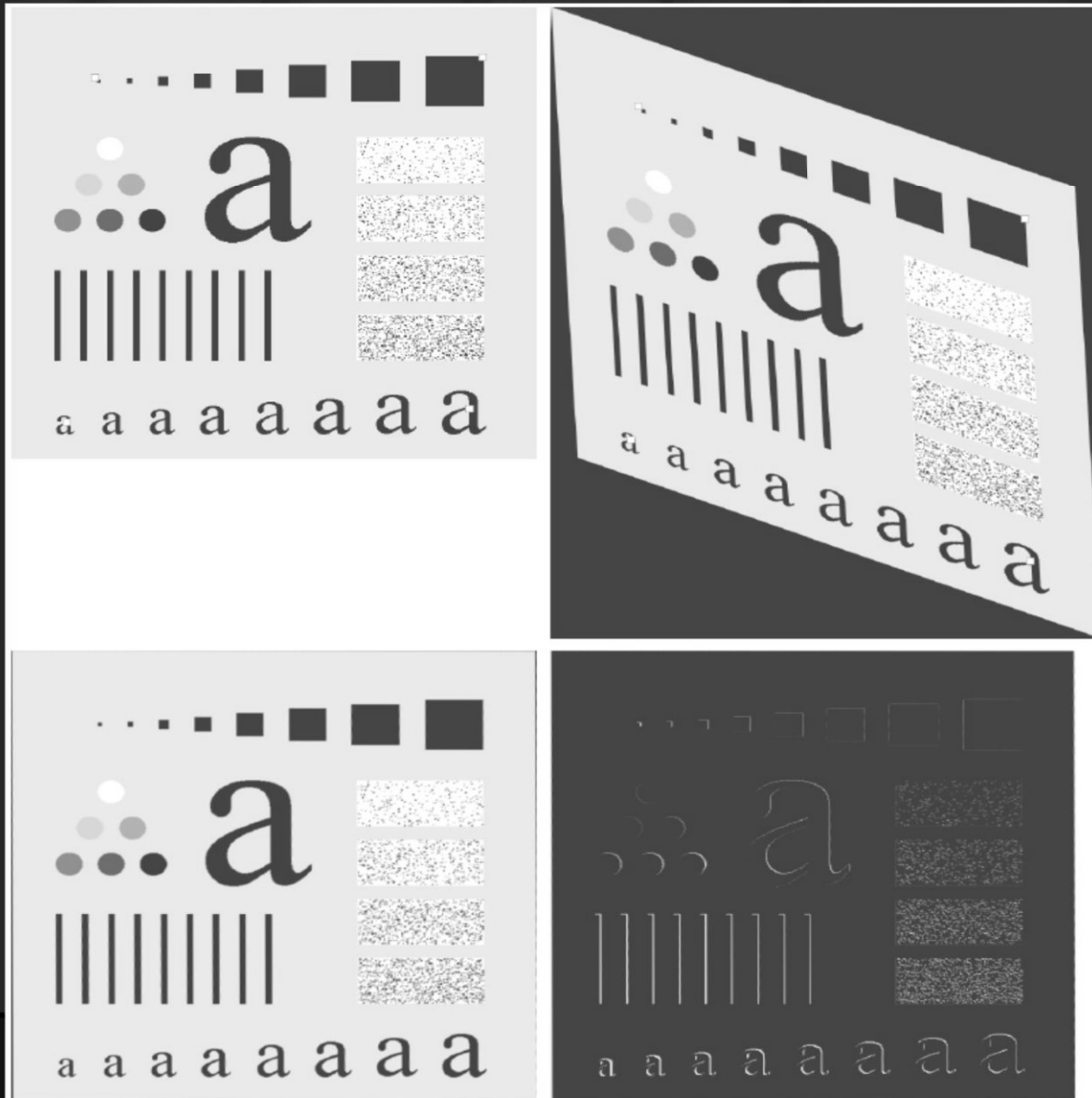


```
clear;  
clc;  
img1=imread('D:\Image processing\IP_sttp\sticks.bmp');  
img2=imread('D:\Image processing\IP_sttp\sticks2.bmp');  
img1=double(img1);  
img2=double(img2);  
imshow(img1);  
figure,imshow(img2);  
final=img1 ./img2;  
final=mat2gray(final);  
imwrite(final,'D:\Image Processing\IP_sttp\sticks3.bmp');  
figure, imshow (final);
```

Image Registration

- To align two or more images of the same scene.
- In case of transformation, form of the transformation function required to achieve a desired geometric transformation was known.
- In image registration, input and output images are known, transformation that produce output image from input image is unknown.
- The problem is to estimate transformation function and use it to register two images.
- Input image is the image which we wish to transform, and what we call the reference image as the image against which we wish to register the input.

Image Registration



a	b
c	d

FIGURE 2.37

Image registration.
 (a) Reference image. (b) Input (geometrically distorted image). Corresponding tie points are shown as small white squares near the corners.
 (c) Registered image (note the errors in the borders).
 (d) Difference between (a) and (c), showing more registration errors.